



ARRAY ASSIGNMENTS

SECTION A: CONDITIONAL REPLACE (5 Assignments)

Assignment 1.1

Write a C program to replace all even numbers with 1 and all odd numbers with 0 in an array: {12, 7, 18, 9, 24, 11, 30}.

Assignment 1.2

Write a C program to replace all multiples of 4 with 0 in an array: {8, 15, 12, 7, 16, 20, 9}.

Assignment 1.3

Write a C program to replace all multiples of 3 with 0 and all multiples of 2 with 1 in an array: {9, 10, 15, 8, 12, 25, 30}. (If multiple of both, replace with 1)

Assignment 1.4

Write a C program to replace all prime numbers with -1 in an array: {7, 12, 13, 18, 11, 20, 9}.

Assignment 1.5

Write a C program to replace all duplicate numbers with -1 in an array: {10, 15, 10, 20, 15, 30}.

SECTION B: CONDITIONAL DELETION (4 Assignments)

Assignment 1.6

Write a C program to delete all odd numbers from an array: {12, 7, 18, 9, 24, 11, 30}.

Assignment 1.7

Write a C program to delete all multiples of 4 from an array: {12, 15, 8, 9, 20, 7, 30}.

Assignment 1.8

Write a C program to delete all prime numbers from an array: {7, 12, 13, 18, 11, 20, 9}.

Assignment 1.9

Write a C program to delete the smallest element from an array: {12, 7, 18, 5, 22, 9, 14}.

SECTION C: SKIP & DELETION OPERATIONS (3 Assignments)

Assignment 1.10

Write a C program to delete all elements that come immediately AFTER an odd number in an array: {5, 10, 8, 15, 12, 7, 20}.

Assignment 1.11

Write a C program to Skip Two elements after every multiple of 3 in an array: {3, 10, 15, 8, 20, 9, 25, 12}.

Assignment 1.12

Write a C program to Delete TWO elements after every multiple of 5 in an array: {5, 10, 7, 8, 20, 7, 25, 12, 30}.

SECTION D: SORTING (3 Assignments)

Assignment 1.13

Write a C program to sort an array in ascending order (without using built-in functions). Array: {5, 2, 8, 1, 9, 3}.

Assignment 1.14

Write a C program to sort an array in descending order. Array: {12, 5, 18, 3, 20, 8}.

Assignment 1.15

Write a C program to sort an array, then count the number of unique elements. Array: {4, 5, 4, 6, 6, 7, 8}.

SECTION E: FREQUENCY, UNIQUE, DUPLICATE (5 Assignments)

Assignment 1.16

Write a C program to count the frequency of each element in an array: {4, 5, 4, 6, 6, 7, 8}.

Assignment 1.17

Write a C program to find and print all unique elements in an array (elements that appear only once): {10, 15, 10, 20, 15, 30}.

Assignment 1.18

Write a C program to identify and print all duplicate elements in an array: {10, 15, 10, 20, 15, 30, 10}.

Assignment 1.19

Write a C program to delete all duplicate elements (keep only first occurrence): {10, 15, 10, 20, 15, 30, 10}.

Assignment 1.20

Write a C program to delete all unique elements from an array (keep only duplicate values): {10, 15, 10, 20, 15, 30, 25}.
